Predictive Churn Analysis in Telecommunications

A Multi-Model Approach to Predictive Modeling and Clustering

Azeezat Bello
CSC 4740
Georgia State University

Abstract

Customer churn is a significant challenge for companies reliant on recurring revenue. This project utilizes the Telco Customer Churn dataset to predict churn likelihood and analyze behavioral differences across customer lifecycle stages. By implementing a pipeline of data cleaning, feature encoding, K-Means clustering, and classification modeling, we identified key drivers of churn such as tenure and contract type. Results indicate that Logistic Regression provides the most balanced predictive performance, while clustering reveals high-risk segments with churn rates as high as 41.68%.

1. Introduction

Customer churn refers to the event where a customer stops using a service or cancels their subscription. It is a major concern for companies that rely on recurring revenue, such as telecommunications and subscription-based platforms. Retaining existing customers is often more cost-effective than acquiring new ones, which makes understanding and predicting churn an important problem in data mining and business analytics.

Machine learning provides useful tools for identifying patterns in customer behavior and estimating the likelihood that a customer will leave. By analyzing features such as tenure, contract type, service usage, and billing information, it is possible to detect trends that are associated with higher churn risk. However, churn is not uniform across all customers. Behavior varies depending on factors such as how long a customer has been with the service and what type of plan they are on.

This project focuses on analyzing customer churn using both predictive modeling and clustering techniques. The goal is not only to predict whether a customer will churn, but also to understand how churn behavior differs across customer groups and lifecycle stages. This approach allows for both accurate prediction and deeper insight into why customers leave.

The dataset used in this project is the Telco Customer Churn dataset, which contains demographic information, service details, billing data, and churn status for each customer. The overall system follows a structured pipeline: data cleaning, feature encoding, exploratory data analysis, clustering, model training, and evaluation.

To achieve the project goals, several key questions are addressed:

- Can meaningful customer groups be identified using clustering techniques?
- Do different customer groups have different churn rates?
- How does churn behavior change across lifecycle stages?
- How accurately can machine learning models predict churn?
- Which features have the strongest influence on churn?

By combining clustering with classification models, this project provides both predictive results and an explanation of customer behavior. The final outcome is a system that can estimate churn risk while also identifying the type of customer being analyzed, which supports more informed decision-making.

2. Methodology

This project follows a structured pipeline that combines data preprocessing, exploratory analysis, clustering, and predictive modeling. The goal is to both estimate churn risk and understand how customer behavior differs across groups and lifecycle stages.

2.1 Data Preprocessing

The Telco Customer Churn dataset was first cleaned to ensure consistency and usability for modeling. The TotalCharges column was originally stored as a string and contained invalid entries. It was converted to a numeric format, and rows with missing values were removed. This resulted in a cleaned dataset with minimal data loss.

Categorical variables were then transformed into numeric form using one-hot encoding. The drop_first=True setting was applied to avoid redundant features. This step allowed all machine learning models to process the data effectively.

The target variable, Churn, was converted into binary form, where 1 represents customers who churned and 0 represents customers who did not.

2.2 Feature Preparation and Splitting

After preprocessing, the dataset was divided into input features (X) and the target variable (y). The data was then split into training and testing sets using an 80/20 ratio. Stratified sampling was used to preserve the original churn distribution in both sets, ensuring a fair evaluation.

Feature scaling was applied using StandardScaler for models that are sensitive to feature magnitude, specifically Logistic Regression and K-Means clustering. Tree-based models such as Decision Tree and Random Forest were trained on the unscaled data, as they are not affected by feature scale.

2.3 Predictive Modeling

Three classification models were selected to predict customer churn:

- **Logistic Regression** was used as a baseline model due to its simplicity and interpretability. It provides insight into how different features influence the probability of churn.
- **Decision Tree** was used to capture rule-based relationships in the data. This model helps identify clear decision paths that lead to churn, making it useful for interpretation.
- **Random Forest** was used to improve predictive performance by combining multiple decision trees. This model reduces overfitting and provides feature importance scores for further analysis.

Each model was trained on the training dataset and evaluated on the test dataset. The following metrics were used for evaluation:

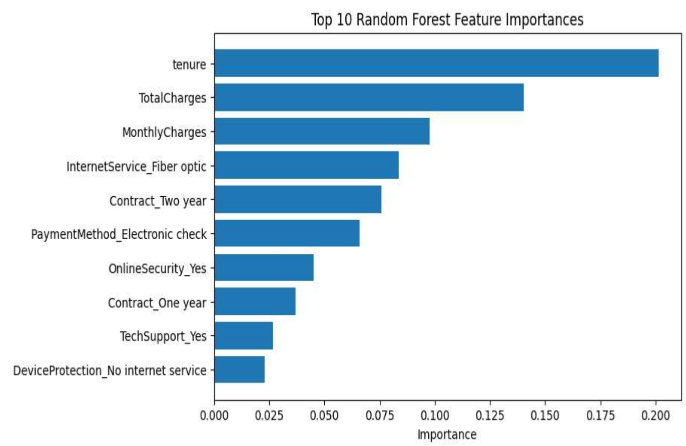
- Accuracy
- Precision
- Recall
- F1 Score

These metrics provide a balanced view of model performance, especially given the class imbalance in the dataset.

2.4 Feature Importance Analysis

To better understand which factors influence churn, feature importance was analyzed using the Random Forest model. This method ranks features based on their contribution to prediction performance.

This step allows the model results to be connected back to the patterns observed during the exploratory data analysis phase, ensuring consistency between statistical insights and predictive modeling. The Top 10 Results of the Random Forest Feature Analaysis are as displayed:



2.5 K-Means Clustering

In addition to predictive modeling, K-Means clustering was applied to identify customer segments. The clustering was performed using selected features that capture customer behavior, including:

- tenure
- MonthlyCharges
- TotalCharges
- SeniorCitizen

These features were scaled before clustering to ensure fair distance calculations.

The number of clusters was set to three to create interpretable customer groups. Each cluster was then analyzed based on its average feature values and churn rate. This allowed the identification of distinct customer types, such as long-term stable customers, newer customers, and higher-risk groups.

2.6 System Overview

The overall system can be summarized as the following pipeline:

- Dataset
- Data Cleaning
- Feature Encoding
- Exploratory Analysis
- Clustering
- Train/Test Split
- Model Training
- Evaluation
- Analysis

This design ensures that the project addresses both prediction and interpretation. The classification models provide churn predictions, while clustering adds a segmentation layer that explains differences in customer behavior across groups.

3. Evaluation

This section evaluates the performance of the classification models and analyzes the results of clustering. The goal is to assess how well the models predict customer churn and to identify meaningful patterns that explain customer behavior.

3.1 Experimental Setup

The dataset was split into training and testing sets using an 80/20 ratio, with stratified sampling to preserve the original class distribution. All models were trained on the training set and evaluated on the test set to ensure a fair assessment of performance.

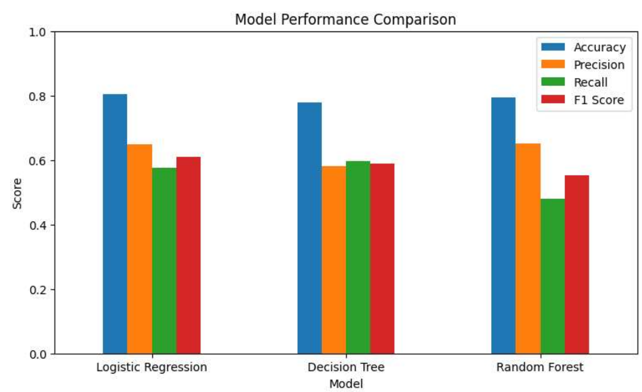
For classification, three models were tested: Logistic Regression, Decision Tree, and Random Forest. Each model was evaluated using accuracy, precision, recall, and F1 score. These metrics were chosen to provide a balanced evaluation, especially since the dataset contains more non-churned customers than churned customers.

For clustering, K-Means was applied using selected behavioral features, and the resulting groups were evaluated based on their churn rates and average feature values.

3.2 Model Performance

The results of the three classification models are summarized below:

	Logistic Regression	Decision Tree	Random Forest
Accuracy	0.8038	0.7783	0.7932
Precision	0.6476	0.5807	0.6509
Recall	0.5749	0.5963	0.4786
F1-Score	0.6091	0.5884	0.5516



Among the three models, Logistic Regression achieved the best overall performance, with the highest accuracy and F1 score. It also maintained a strong balance between precision and recall, making it the most reliable model for predicting churn.

The Decision Tree model showed slightly lower accuracy but had comparable recall, indicating that it was reasonably effective at identifying churn cases, though with more false positives.

The Random Forest model achieved strong precision but lower recall, meaning it was more conservative in predicting churn and tended to miss a larger portion of actual churn cases. While its predictions were often correct when identifying churn, it was less effective at capturing all at-risk customers.

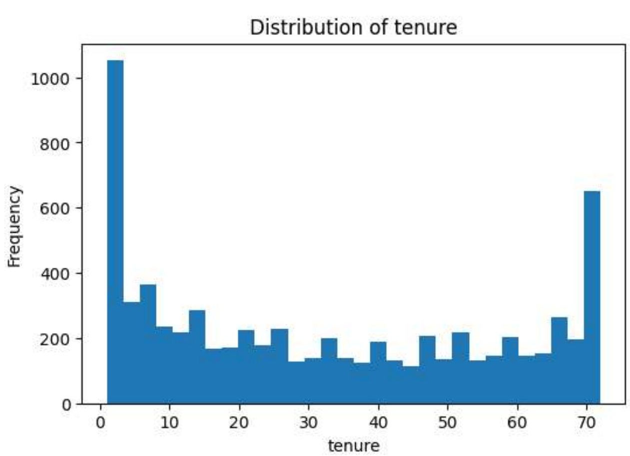
Overall, Logistic Regression provided the most balanced and consistent performance across all evaluation metrics.

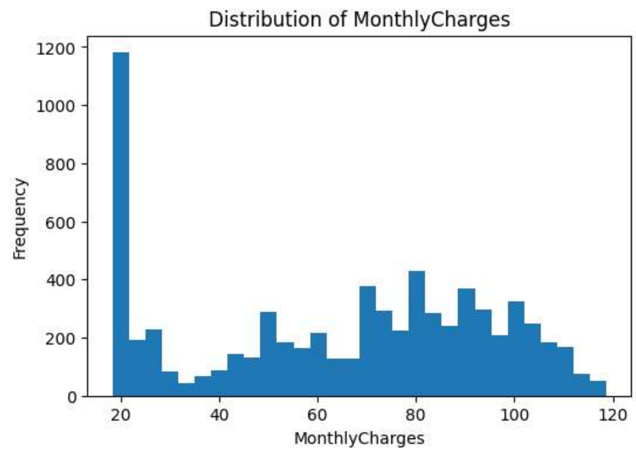
3.3 Feature Importance and Interpretation

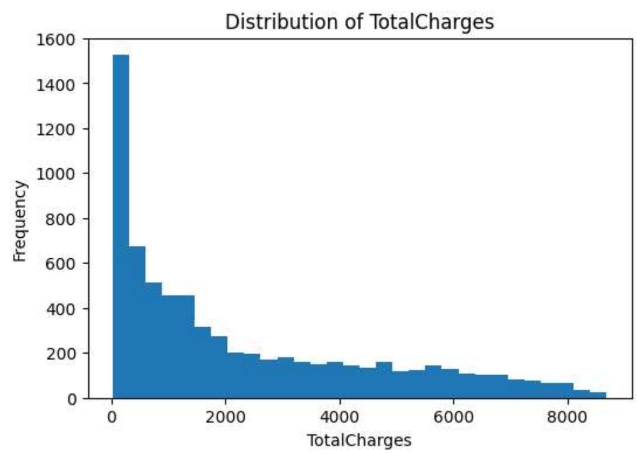
Feature importance analysis from the Random Forest model revealed that the most influential variables in predicting churn include:

- Tenure
- TotalCharges
- MonthlyCharges
- InternetService (Fiber optic)
- Contract type
- Payment method (Electronic check)

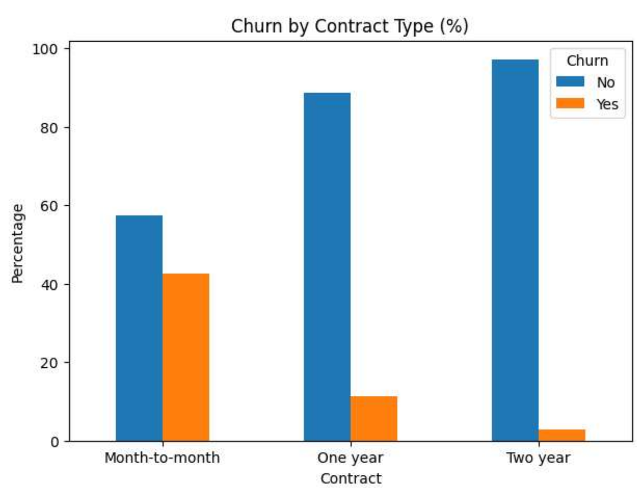
These results align closely with the patterns observed during the exploratory data analysis phase. Customers with shorter tenure were significantly more likely to churn, indicating that early-stage customers are less stable. Higher monthly charges were also associated with increased churn risk, suggesting that pricing plays a role in customer dissatisfaction, however there is the nuance in that higher charges are not necessarily factorial in causing churn if the product lives up to the user expectations, combing this with TotalCharges gives a much better picture. The distirbution for the features of tenure, MonthlyCharges and TotalCharges is displayed below:







Contract type was another key factor, with long-term contracts strongly reducing the likelihood of churn. This reinforces the idea that customer commitment and service duration are critical drivers of retention. The results from the EDA are displayed below.

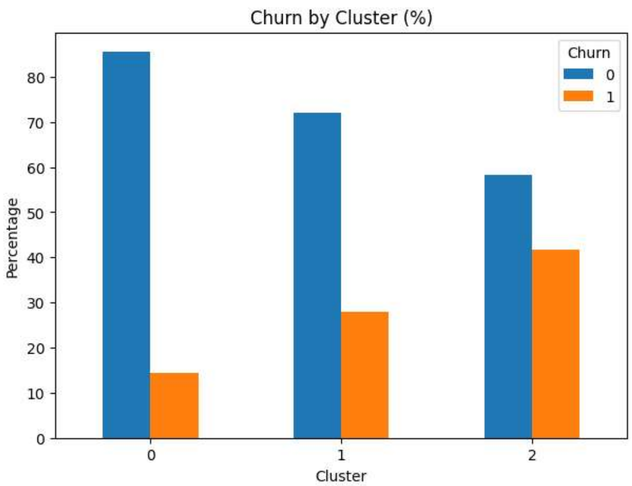


3.4 Clustering Results

K-Means clustering was used to identify customer segments based on key behavioral features. The model was configured with three clusters to create interpretable groups.

The churn rates across clusters were:

Cluster 0	Cluster 1	Cluster 2
14.45% Churn	27.95% Churn	41.68% Churn



These results show a clear separation between customer groups, with Cluster 2 representing the highest-risk segment.

Further analysis of cluster characteristics revealed:

- **Cluster 0**: Customers with long tenure, high total charges, and low churn rates. This group represents stable, long-term customers.
- **Cluster 1**: Customers with shorter tenure and lower total charges. This group represents newer customers with moderate churn risk.
- **Cluster 2**: Customers with moderate tenure, higher monthly charges, and the highest churn rate. This group represents the most at-risk segment.

These findings confirm that customer behavior differs significantly across groups and that churn risk is not uniform across the dataset.

3.5 Key Observations

The evaluation results highlight several important insights:

- Churn is strongly influenced by customer lifecycle stage, with newer customers being more likely to leave.
- Pricing-related factors, such as monthly charges, play a significant role in churn behavior.
- Long-term contracts significantly reduce churn risk, indicating the importance of customer commitment.
- Machine learning models can predict churn with reasonable accuracy, with Logistic Regression providing the most balanced performance.
- Clustering reveals distinct customer segments, allowing churn behavior to be analyzed beyond simple prediction.

Overall, the combination of predictive modeling and clustering provides both accurate churn estimation and meaningful insight into customer behavior.

4. Learning Outcomes

This project involved multiple stages, including data preprocessing, exploratory data analysis, clustering, predictive modeling, and result interpretation. Each stage contributed to a deeper understanding of both the technical and analytical aspects of data mining.

The data preparation phase highlighted the importance of properly cleaning and structuring data before applying any models. Issues such as incorrect data types and missing values had to be addressed early, as they can directly affect the quality of the results. This reinforced the idea that preprocessing is a critical step in any data-driven project.

During the exploratory data analysis phase, patterns related to churn behavior began to emerge. Observations such as higher churn among customers with shorter tenure and month-to-month contracts helped guide the direction of the modeling phase. This showed how early analysis can inform later decisions and improve the overall approach.

The modeling phase provided practical experience with different machine learning techniques. Logistic Regression, Decision Tree, and Random Forest were implemented and compared using multiple evaluation metrics. This process demonstrated that no single model is universally best, and that performance must be evaluated based on the specific problem and dataset. In this case, Logistic Regression provided the most balanced performance.

Clustering added another layer of understanding by grouping customers into segments based on their characteristics. This showed that churn is not uniform across all customers, but varies depending on factors such as tenure and billing behavior. Combining clustering with predictive modeling helped provide both accurate predictions and meaningful insights.

One challenge encountered during the project was ensuring that all preprocessing steps were applied consistently across different stages. This required careful organization of the workflow to avoid errors when transitioning from analysis to modeling. Additionally, interpreting model results required attention to multiple evaluation metrics, rather than relying on a single value such as accuracy.

Overall, this project improved understanding of the full data mining pipeline, from raw data to final insights. It also reinforced the importance of combining different techniques to gain both predictive and analytical value from a dataset.

5. Group Member Evaluation

Since this project was completed individually, there are no additional group members to evaluate. However, a self-evaluation is provided based on effort, participation, and contribution.

- **Effort:9/10**
  A strong level of effort was maintained throughout the project, especially during the data analysis and modeling phases. Time was invested in understanding the dataset and ensuring that each step was implemented correctly.
- **Participation: 9/10**
  All stages of the project were completed, including preprocessing, exploratory analysis, clustering, model training, and evaluation. Active engagement was maintained across the entire workflow.
- **Contribution: 9/10**
  The full system, including both predictive modeling and clustering, was successfully implemented. The project meets its original goals of analyzing churn through both lifecycle and segmentation perspectives.
- **Comments:**
  While the project was completed successfully, time management could be improved in future work. Some stages took longer than expected, which required working under tighter time constraints later in the project. Overall, the experience was valuable and reinforced the importance of planning and pacing across all phases.

6. Demo Video Access

The demo video for this project can be accessed using the link below:

https://drive.google.com/file/d/1wg0S1LokJvRhQ9zFcXR_DX3-e1m0ltUF/view?usp=sharing

The demo presents the full workflow of the system, including data preprocessing, clustering, model training, and churn prediction. It also demonstrates how the final system can be used to input customer data and generate predictions along with customer segmentation insights.